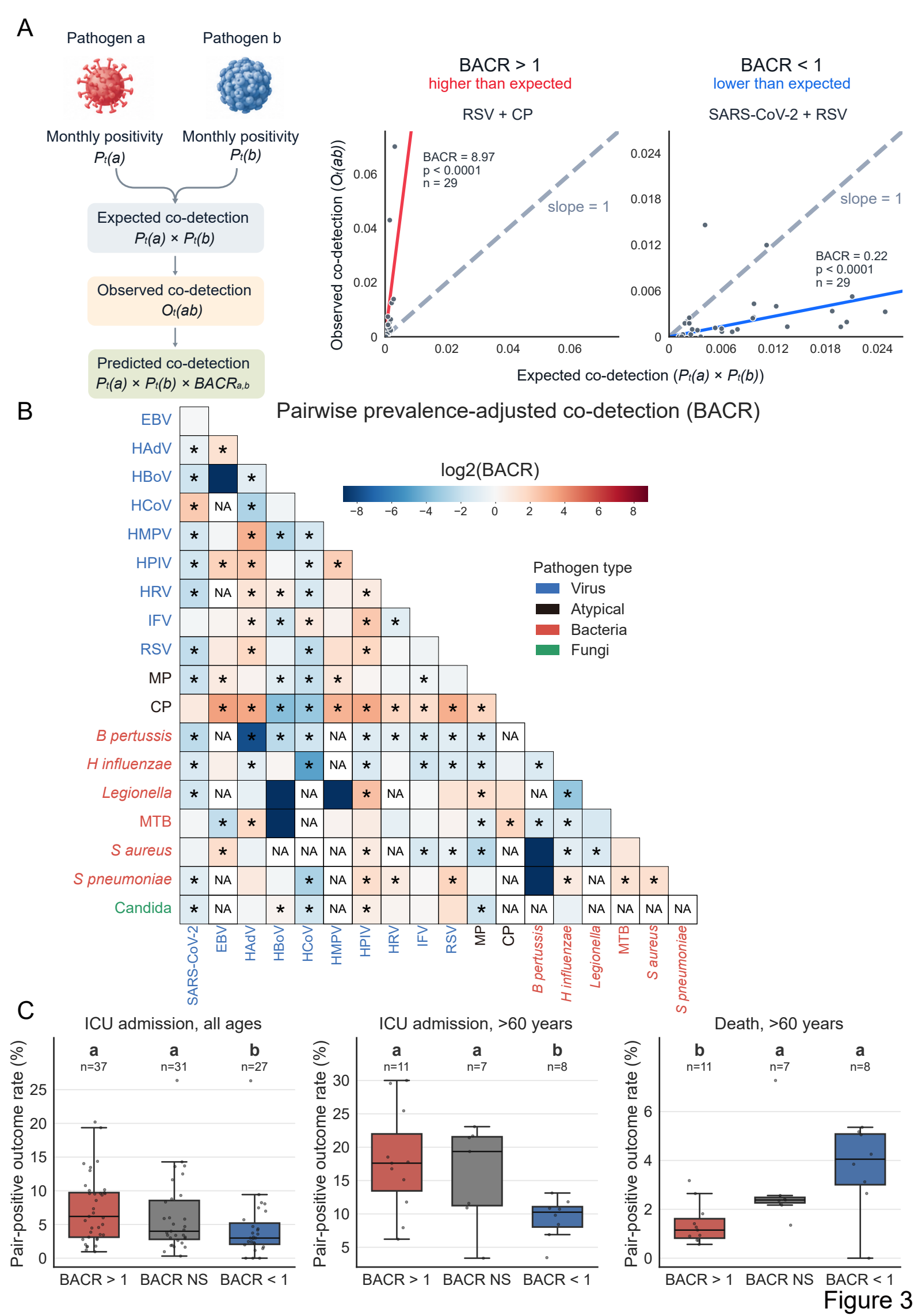

A
Pathogen a
Pathogen b
Monthly positivity
$P_t(a)$
Monthly positivity
$P_t(b)$
Expected co-detection
$P_t(a) \times P_t(b)$
Observed co-detection
$O_t(ab)$
Predicted co-detection
$P_t(a) \times P_t(b) \times BACR_{a,b}$
BACR > 1
higher than expected
BACR < 1
lower than expected
RSV + CP
SARS-CoV-2 + RSV
BACR = 8.97
p < 0.0001
n = 29
slope = 1
BACR = 0.22
p < 0.0001
n = 29
Observed co-detection ($O_t(ab)$)
Expected co-detection ($P_t(a) \times P_t(b)$)
B
Pairwise prevalence-adjusted co-detection (BACR)
log2(BACR)
Pathogen type
Virus
Atypical
Bacteria
Fungi
EBV
HAdV
HBoV
HCoV
HMPV
HPIV
HRV
IFV
RSV
MP
CP
B pertussis
H influenzae
Legionella
MTB
S aureus
S pneumoniae
Candida
SARS-CoV-2
C
ICU admission, all ages
ICU admission, >60 years
Death, >60 years
Pair-positive outcome rate (%)
a n=37
a n=31
b n=27
a n=11
a n=7
b n=8
b n=11
a n=7
a n=8
BACR > 1
BACR NS
BACR < 1

Figure 3